ECG Arrhythmia Classification

A DATA-AUDITED PIPELINE FOR FIVE-CLASS ECG BEAT CLASSIFICATION USING THE MIT-BIH ARRHYTHMIA DATABASE

Student present :

Mohamad Amr Zaiter – Hasan Haider

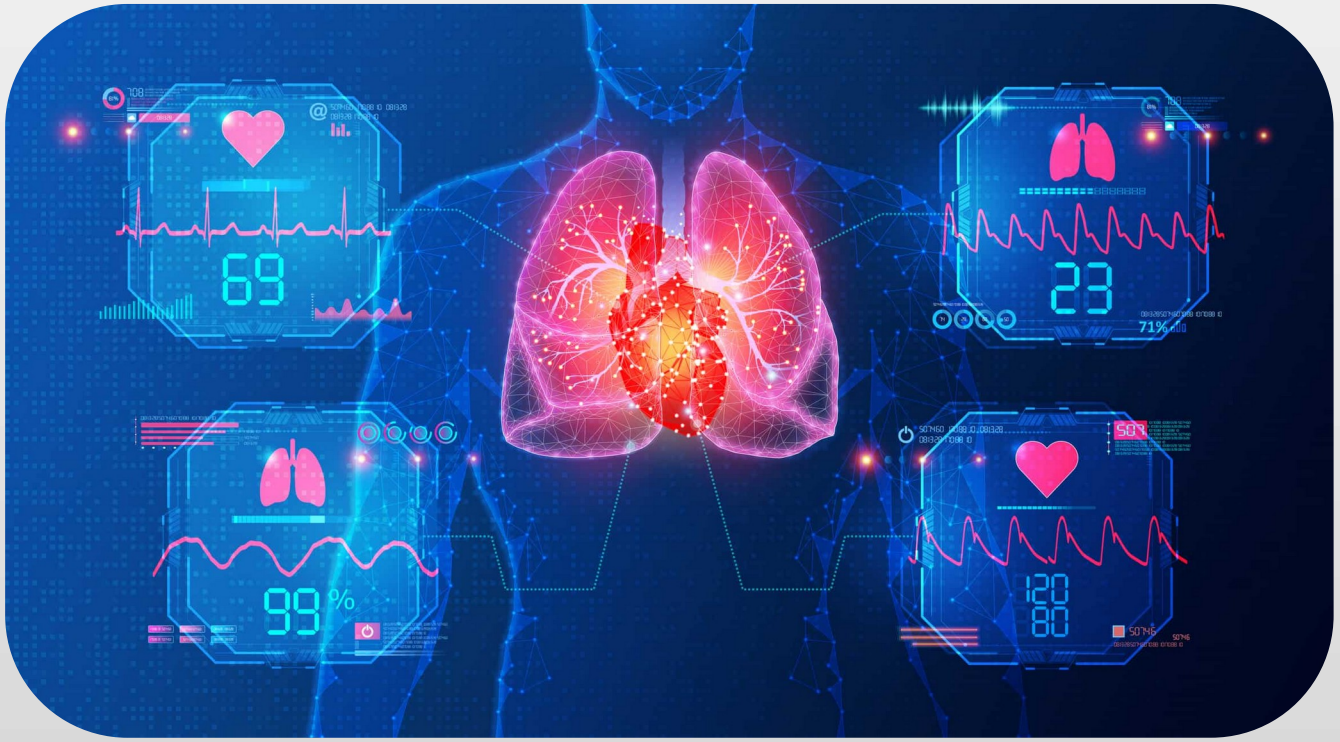
Supervisors :

Dr. Medin Abd Alhamed · Eng. Toka Alzoebi



Project Overview :

- A reproducible pipeline for preparing MIT-BIH ECG beats for five-class AAMI arrhythmia classification, from data auditing and patient-level splitting to signal filtering and feature extraction.



Problem & Approach

Problem

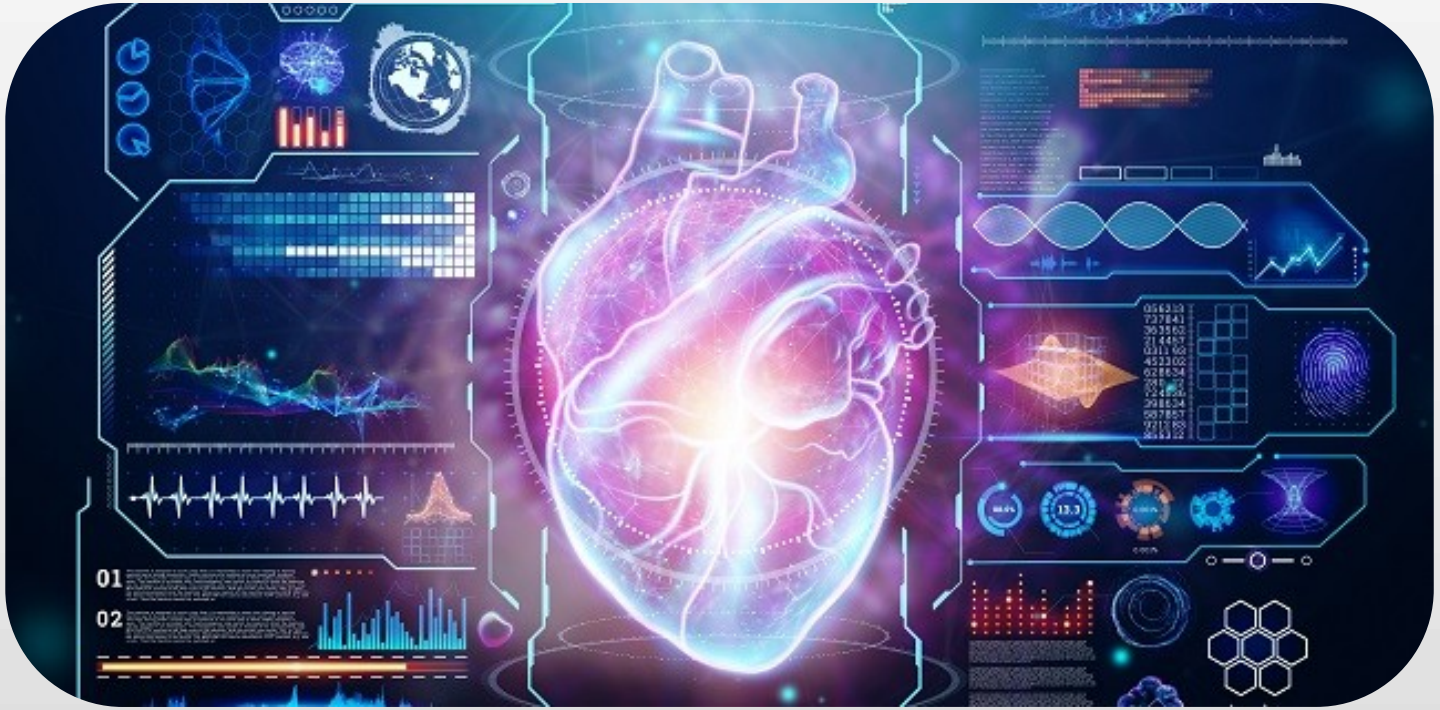
- ECG annotations include non-beat events, rare classes, and patient-level dependencies.

Plan

- Verified pipeline: AAMI mapping, patient-wise splits, and ECG preprocessing.

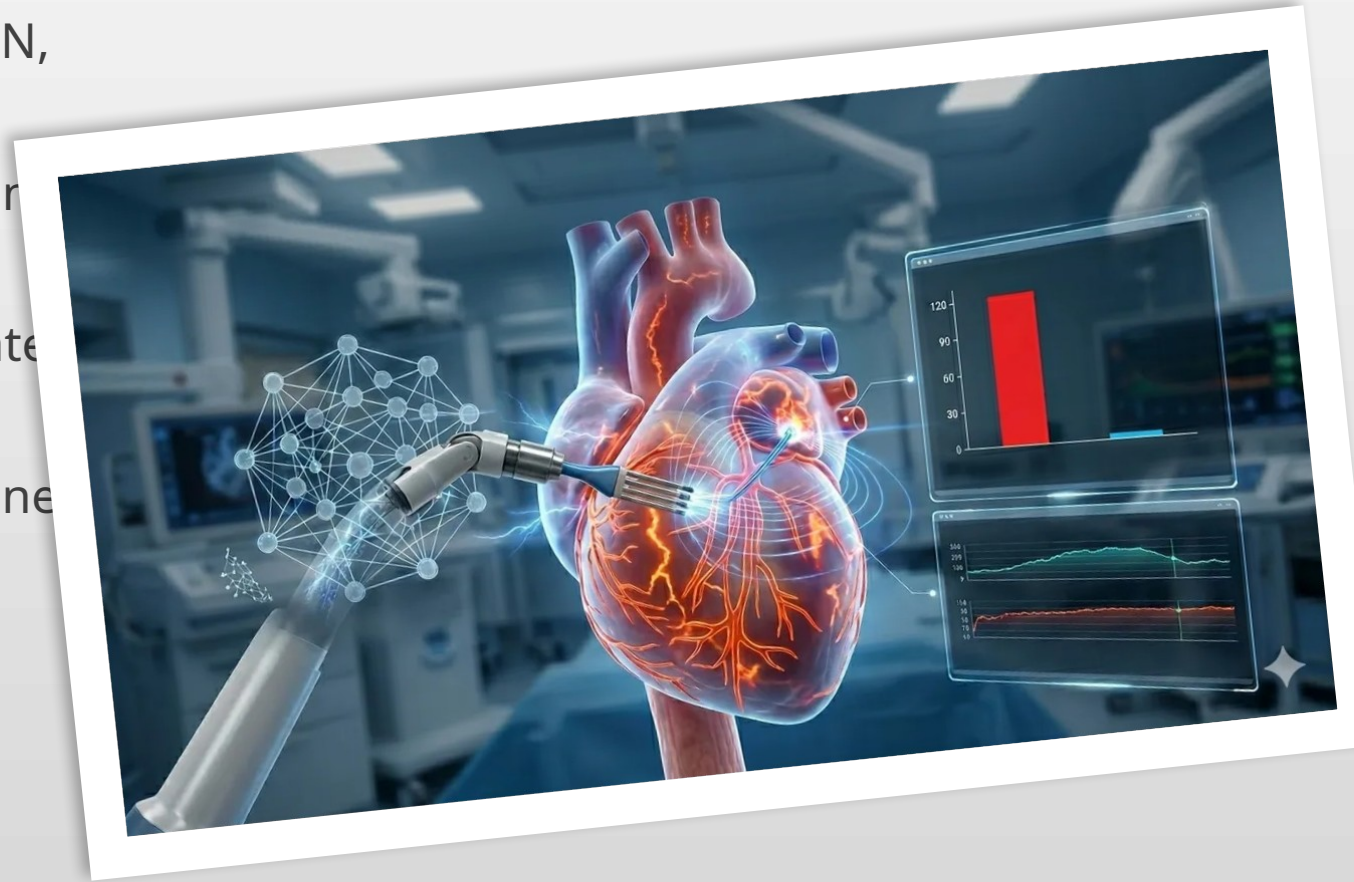
Project Significance

- Builds a traceable foundation for ECG beat classification.
- Uses expert annotations and standard AAMI classes for consistent labels.
- Separates patients across train, validation, and test to reduce data leakage.
- Checks signal quality and class-wise retention before model training.



Project Objectives

- Audit the MIT-BIH database and verify source-file integrity.
- Map annotated beats into the five AAMI classes: N, S, V, F, and Q.
- Create and freeze patient-level training, validation, and test splits.
- Evaluate ECG filter candidates and select a validated preprocessing setup.
- Extract eight beat-level features and verify baseline model readiness.



Reference Studies & Data Sources

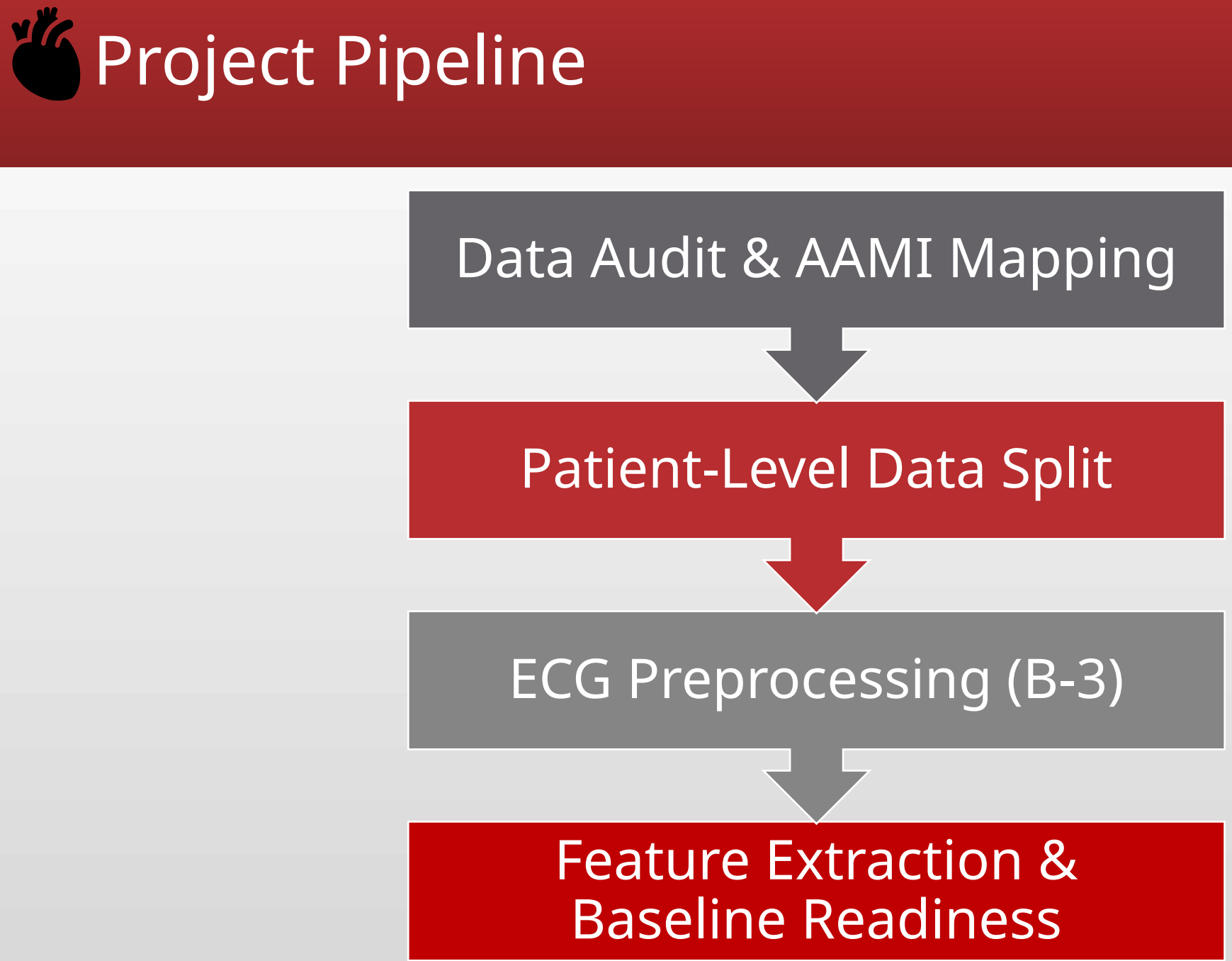
CORE REFERENCES & PROJECT DATA SOURCES				
REFERENCE	**SCOPE**	**ROLE IN PROJECT**	**USE IN THIS WORK**	**EVIDENCE**
de Chazal et al. (2004)	ECG beat classification	Morphology + RR interval features	AAMI mapping and DS1 / DS2 protocol	Exact count reproduction
ANSI/AAMI EC57:1998	ECG algorithm evaluation	Standard class grouping	Five-class scheme: N / S / V / F / Q	Schema basis
PhysioNet MIT-BIH Arrhythmia Database v1.0.0	48 ECG records; 47 patients	Expert beat annotations	Primary dataset (360 Hz; 2 leads)	209 / 209 files SHA-256 verified
Luo & Johnston (2010)	ECG filtering review	Filter design and trade-offs	Filtering context for pilot rationale	Background reference

Dataset Overview :

- ✓ MIT-BIH Arrhythmia Database: 48 records from 47 patients, sampled at 360 Hz across two ECG leads.
- ✓ A total of 112,647 annotation events were reviewed; 109,494 classifiable beats were mapped to five AAMI classes.
- ✓ The audit excluded 3,153 non-beat annotations from classification.
- ✓ All 209 source files passed SHA-256 integrity verification.

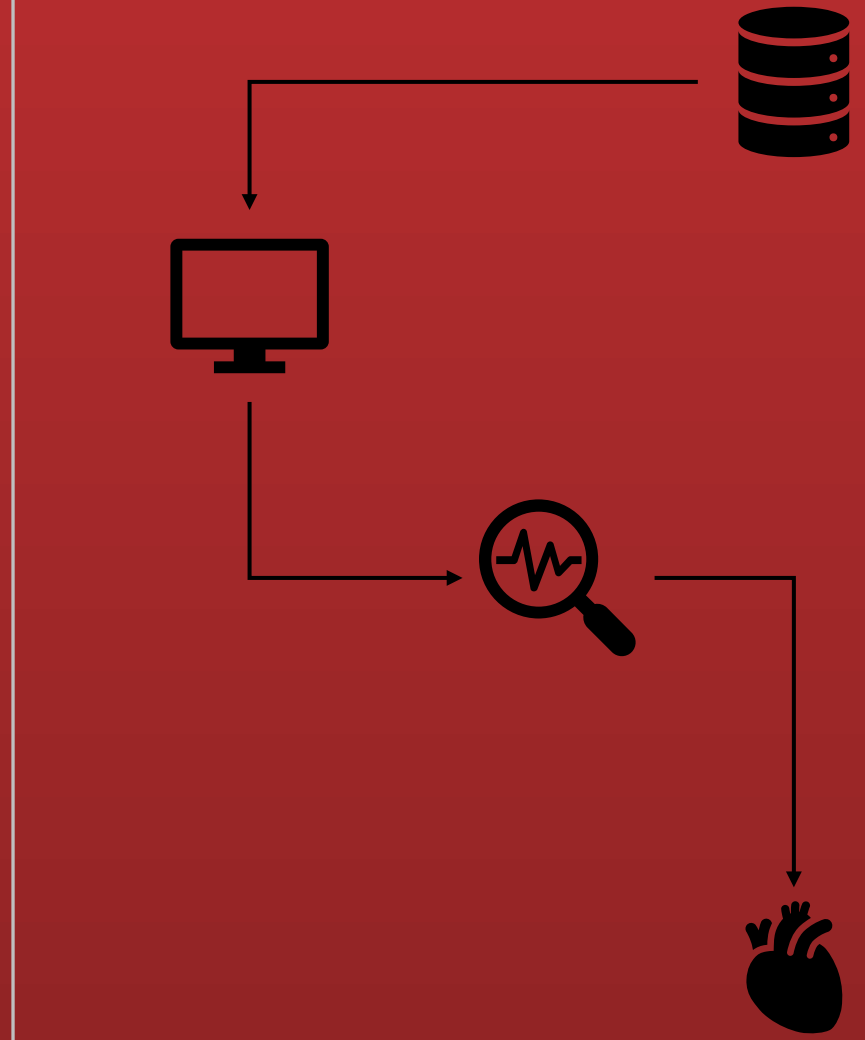


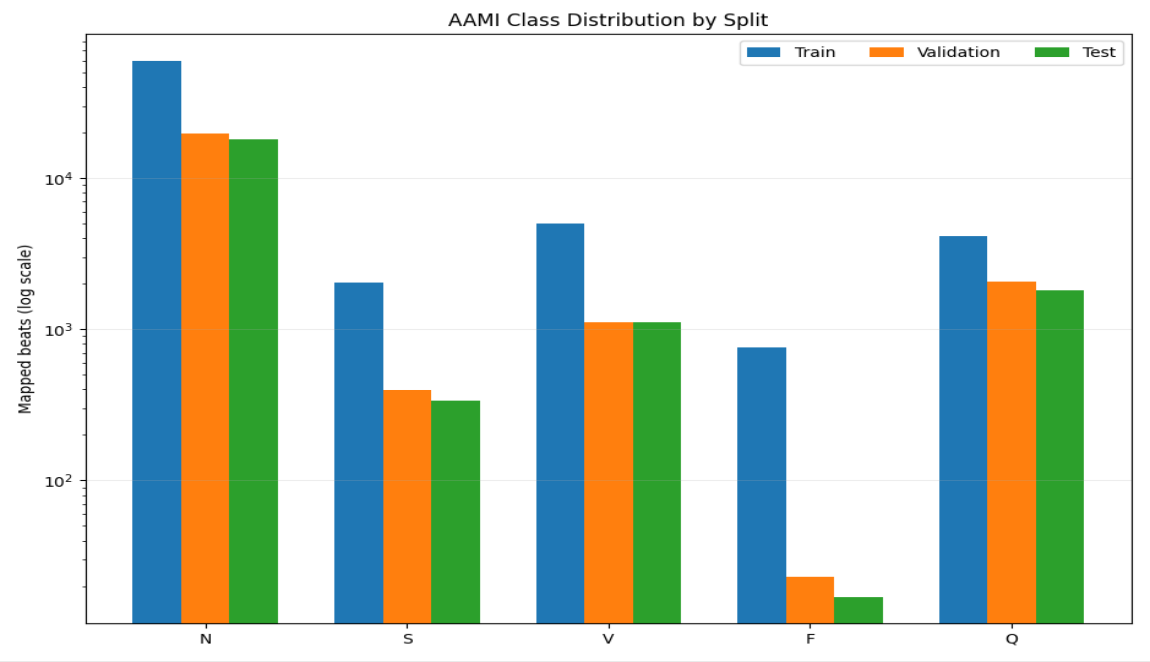
Project Pipeline

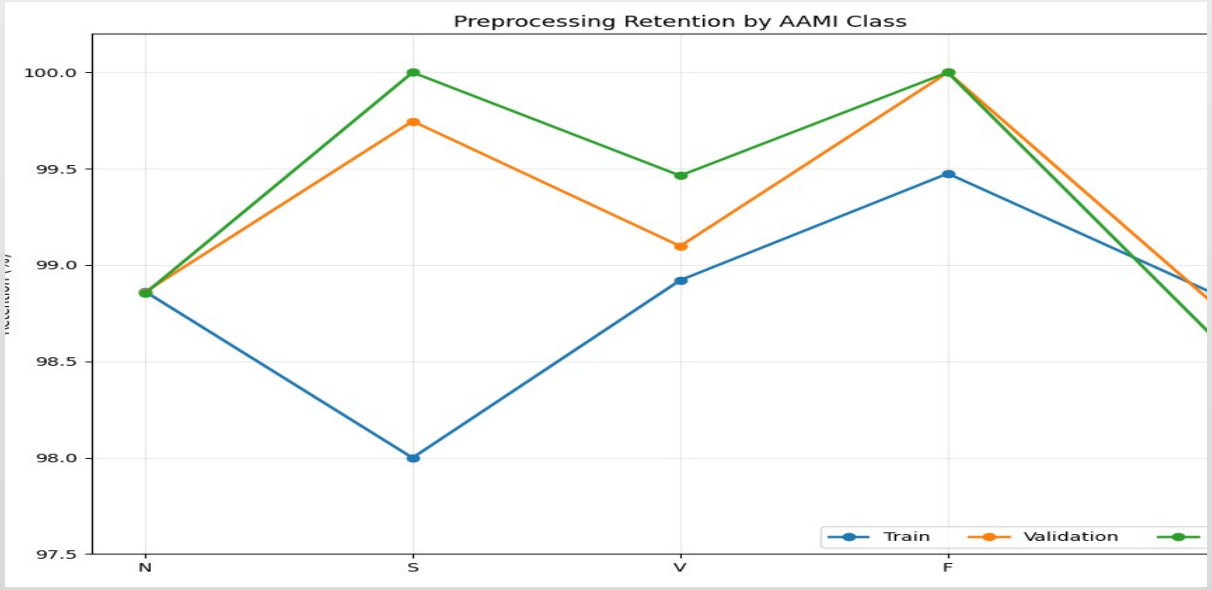


Validation Method :

- Annotation counts were cross-checked against official reference tables.
- Patient-level splits were frozen and verified against cross-split leakage.
- Eight filter candidates were tested against predefined signal-quality criteria.
- Retention was enforced at ≥95% in every class-by-split cell; 100 unit tests and 196 independent checks passed.







Results

Mapped beats : 109,494

Retained beats : 108,192 (98.81%)

Unit tests : 100 / 100 PASS

Independent checks : 196 / 196 PASS

1 DATA INTEGRITY

RAW FILES

209 / 209 verified

UNIT TESTS

100 / 100 PASS

INDEPENDENT CHECKS

196 / 196 PASS

2 ANNOTATION AUDIT

ANNOTATION EVENTS

112,647

CLASSIFIABLE BEATS

109,494

NON-BEAT EVENTS EXCLUDED

3,153

3 PATIENT-LEVEL SPLIT

TRAINING

34 records | 33 patients

VALIDATION

7 records | 7 patients

TESTING

7 records | 7 patients

4 PREPROCESSING

SELECTED FILTER

B-3 | Butterworth-4

RETAINED BEATS

108,192

MINIMUM CLASS × SPLIT RETENTION

97.999%

Baseline training and model-performance evaluation are not yet completed.

Project Progress

Stage	Main Output	Verified Result	Status
Data acquisition	MIT-BIH raw archive	209 / 209 files SHA-256 verified	Complete
Data audit	Annotation audit	112,647 events; 109,494 beats	Complete
AAMI mapping	Five-class labels	N / S / V / F / Q	Complete
Data splitting	Patient-level split	34 / 7 / 7 records; no overlap	Frozen
Filter pilot	8 filter candidates	B-3 passed all predefined criteria	Selected
Full preprocessing	Filtered beat set	108,192 retained; minimum 97.999%	Complete
Feature schema	Baseline input	8 predictors; 41 / 41 checks	Ready

THANK YOU

